

Swapnil Gautam

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SUMMARY

Machine Learning Engineer with 5+ years of experience building and deploying **production computer vision and multimodal AI systems** for real-world environments. Expertise in **computer vision, deep learning, VLMs, LLMs, RAG, and agentic AI**, with hands-on experience developing **detection, segmentation, classification, 3D scene reconstruction, and spatial intelligence** solutions using PyTorch and TensorFlow. Proven track record taking ML systems from **data curation and model development through optimization, cloud/edge deployment, and production monitoring**, including **real-time inference on resource-constrained hardware using ONNX and TensorRT**.

EXPERIENCE

Aethreallegence (Peoria, IL)

Computer Vision Engineer

April 2026 - Present

- Building a **computer vision platform for inventory estimation, anomaly detection, and theft detection**, using live camera feeds to localize, identify, and count products in real-world environments.
- Built a **zero/one-shot visual recognition pipeline using SAM 2 and DINOv3**, generating SKU embeddings from augmented onboarding images and matching scene-level segmentation proposals using cosine similarity.
- Developed a **multi-camera 3D scene reconstruction pipeline** to combine views from multiple cameras, enabling improved **3D spatial understanding** of real-world environments.

Conduit (San Francisco, CA)

Machine Learning Engineer

September 2025 - January 2026

- Designed and deployed an **end-to-end computer vision pipeline for real-time industrial monitoring**, spanning image acquisition, model inference, cloud services, event streaming, and production deployment.
- Deployed and benchmarked **YOLO-based perception models on NVIDIA Jetson Orin NX and GPU servers**, leveraging **ONNX Runtime and TensorRT** to optimize inference latency and throughput for resource-constrained environments.
- Fine-tuned and deployed a **Vision-Language Model (Llama 3.2)** for machine-state detection and action recognition, improving production monitoring accuracy by **45%**.
- Built a **production-grade RAG application using LangChain and planner-based agents** to retrieve and synthesize machine utilization, maintenance, and operational data from **Qdrant**, with an **ACL-aware pipeline supporting 14K+ documents, metadata filtering, vector search, and PII guardrails**.
- Built data curation and preprocessing pipelines for continuously expanding visual datasets, incorporating quality checks, augmentation, and hard-negative samples to improve model robustness.

Wearable Robotic Systems Lab (Hoboken, NJ)

AI Research Assistant

January 2025 - August 2025

- Developed computer vision algorithms for **lumbar spine detection and localization from MRI images**, supporting automated analysis of spinal anatomy.
- Developed a **Transformer-based multimodal model** for gait parameter estimation using camera, 8-sensor piezoelectric insole, and IMU data; benchmarked against GPR and Bi-LSTM models.
- Worked hands-on with **medical and robotic sensing systems**, including data preprocessing, hardware testing, sensor integration, latency evaluation, and experimental validation.

Wizphys AI (Pune, India)

Founding AI Engineer

April 2022 - June 2024

- **Led the end-to-end development of Wizio's production ML stack** for real-time pose estimation, including data collection, **model training, optimization, deployment, and monitoring**.
- Optimized and deployed computer vision models for both Android and iOS, applying **model pruning and quantization to achieve real-time on-device inference while maintaining accuracy**.
- Designed **scalable cloud data pipelines on Firebase** for data collection and continuous model improvement.
- **Led and mentored a team of 4 ML engineers**, driving rapid feature delivery, code reviews, and production deployments, increasing patient retention by 140% in 6 months.

Camweara (Bangalore, India)

Computer Vision Engineer

July 2021 - January 2022

- **Tested and validated** a custom-deployed **image segmentation model** using **PyTorch** and **ONNX**, cutting third-party API dependencies and **reducing costs by 90%**.
- Designed a custom hand-landmark detection model using TensorFlow, **improving landmark localization accuracy and reducing inference latency through quantization**.

OSV Impex (Gandhidham, India)

Product Developer R&D

July 2020 - July 2021

- Built an **Automatic Number Plate Recognition (ANPR) system using a 2-stage object detection algorithm**, achieving 82%, running on multiple camera RTSP streams simultaneously.

- Developed a **facial recognition** attendance system for 1200 people using Dlib, deployed on Jetson Nano.
- Worked on **3D reconstruction** with custom **object detection, tracking, and camera calibration**, improving industrial robotics workflow.

SKILLS

- **Programming languages and Frameworks:** Python, C++, PyTorch, Transformers, Scikit-learn, CUDA, FFmpeg, TensorFlow, AzureML, AWS, Git version control, CI/CD, SQL, MLOps, VLMs, VLAs, TensorFlow Lite, TensorRT, FAISS, Qdrant DB, Quantization, Edge Deployment, Model Optimization, OCR, Image Classification, Segmentation

EDUCATION

Stevens Institute of Technology (Hoboken, NJ)	<i>Aug 2024 - May 2026</i>
Master of Science in Applied Artificial Intelligence	3.8/4
Coursework: Machine Learning, Deep Learning, 3D Computer Vision, Robotics, Big Data	
Narsee Monjee Institute of Management Studies	<i>July 2016 - Aug 2020</i>
Bachelor of Technology in Computer Science (Mechatronics)	

PUBLICATIONS

- [Presented conference paper ‘Versatile Robotic Arm’ at IEEE PuneCon2023](#)
- [Machine Learning in Production](#)
- [Building an AI-Powered Game!](#)
- [Generative AI with Large Language Models](#)
- [Introduction to LangGraph](#)

PROJECTS

[Agentic AI Tutor \(RAG + Vision + LLM\):](#)

- Built an end-to-end AI tutor using **Gemini API**, supporting RAG and non-RAG modes with **hybrid search (BM25 + ANN vector search)** for improved retrieval accuracy.
- Implemented document understanding with **FAISS & Pinecone (vector DB) + SQLite (metadata storage)**, enabling efficient semantic + keyword-based retrieval.
- Designed a modular LLM pipeline for retrieval, summarization, and contextual Q&A with **Next.js-based UI**.